

2013 HKDSE Physics

Paper 1B Structured Questions

Original questions with official answers and English explanations. Correct answers are red; explanations are blue.

Question 1

A rooftop solar water heater heats 1.5 m^3 of water to 80°C during the day. At night, water from the storage tank circulates through radiators to warm the rooms.



Given: density of water = 1000 kg m^{-3} ; specific heat capacity = $4200 \text{ J kg}^{-1} ^\circ\text{C}^{-1}$.

- During transfer of water, 15% of the energy is lost. Find the heat released to the rooms as the water cools to 60°C . (3 marks)
- At night the radiators maintain an average output power of 4.5 kW . For how many hours can this continue until the water reaches 60°C ? (2 marks)
- The actual heat-release rate gradually decreases over this period. Explain why. (1 mark)

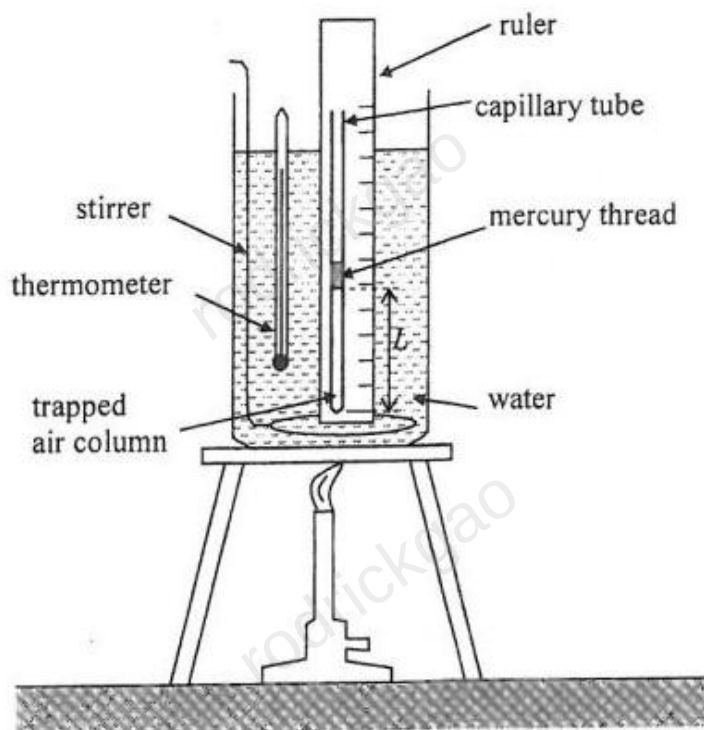
Official marking scheme

- | | | | |
|--------|--|-------------|----------|
| 1. (a) | $(1.5 \times 1000 \text{ kg}) \times 4200 \text{ J kg}^{-1} ^\circ\text{C}^{-1} \times (80 - 60) ^\circ\text{C} \times (1 - 15\%)$
$= 1.07 \times 10^8 \text{ J}$ | 1M+1M
1A | <u>3</u> |
| (b) | $1.07 \times 10^8 \text{ J} \div (4.5 \text{ kW}) \div 3600 \text{ s}$
$= 6.61 \text{ (hours)}$ | 1M
1A | <u>2</u> |
| (c) | Rate of heat transfer drops as the water temperature drops / the room temperature increases / temperature difference drops. | 1A | <u>1</u> |

- (a) Convert volume to mass: $m = \rho V = 1500 \text{ kg}$. Water loses $mc\Delta\theta = 1.26 \times 10^8 \text{ J}$, but only 85% reaches the rooms. Useful heat is $1.071 \times 10^8 \text{ J}$.
- (b) Use useful heat with the radiator output power: $t = Q/P = 23800 \text{ s} = 6.61 \text{ h}$. Keep the unrounded heat in calculation; do not subtract the 15% loss a second time.
- (c) Heat transfer is driven by temperature difference. Cooling water (and warming rooms) reduces that difference, so the instantaneous power falls even though an average power was used in (b).

Question 2

* A small mercury thread traps an air column inside a uniform capillary tube heated in a water bath. Its length L is measured at different temperatures θ .



$\theta / ^\circ\text{C}$	20	92
L / mm	64	80

- (a) Describe the procedures before taking each reading to ensure the trapped air reaches the water temperature. (2 marks)
- (b) Assume L increases linearly with θ throughout.
- (b)(i) Estimate L at 65°C . (2 marks)
- (b)(ii) Find the experimental value of absolute zero. (2 marks)

Official marking scheme

2. (a) Stop heating and stir the water well.

1A

1A 2

(b) (i)
$$\frac{L - 64}{80 - 64} = \frac{65 - 20}{92 - 20}$$
$$L - 64 = 10$$
$$L = 74 \text{ mm}$$

1M

1A 2

(ii) Let $x^\circ\text{C}$ be the absolute zero.

$$\frac{20 - x}{92 - 20} = \frac{64 - 0}{80 - 64}$$

$$20 - x = 288$$

$$x = -268^\circ\text{C}$$

1M

1A 2

- (a) Stop the ongoing temperature rise and stir to remove temperature gradients. Allow the thermometer reading and air-column length to settle before recording.
- (b)(i) The measured gradient is $(80-64)/(92-20) = 2/9$ mm per $^{\circ}\text{C}$. At 65°C , $L = 64 + (2/9)(65-20) = 74$ mm.
- (b)(ii) Extrapolate that measured straight line to $L = 0$: $0 = 64 + (2/9)(x-20)$, giving $x = -268^{\circ}\text{C}$. This is an experimental estimate, not the accepted -273.15°C . Do not force the data to match the accepted value.

Question 3

A lift car of weight 8000 N rises at a constant 2 m s^{-1} . A motor drives a drum with a cable wound around it; the other cable end is fixed to the drum at P (Figure 3.1). Neglect air resistance and cable mass.

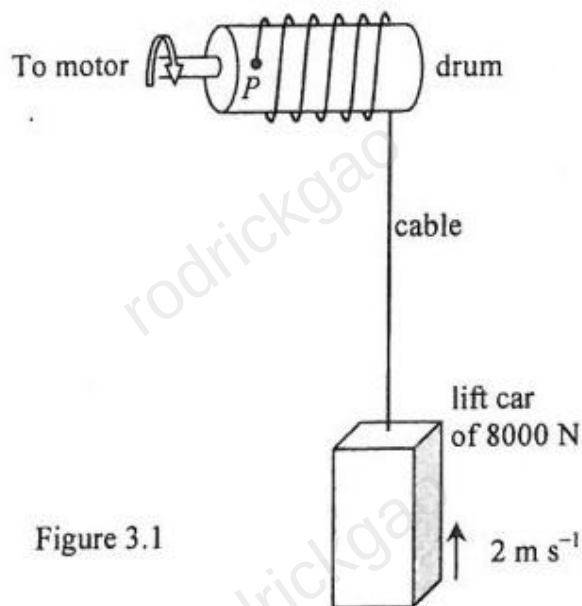


Figure 3.1

- (a)(i) Calculate the mechanical power delivered to the rising car. (2 marks)
- (a)(ii) The motor has total mechanical output 20 kW . Find the power lost overcoming friction between moving parts. (1 mark)
- (b) A 7000 N counterweight is installed at the other end, moving opposite to the car (Figure 3.2). The car again rises at 2 m s^{-1} ; the cable does not slip on the drum.

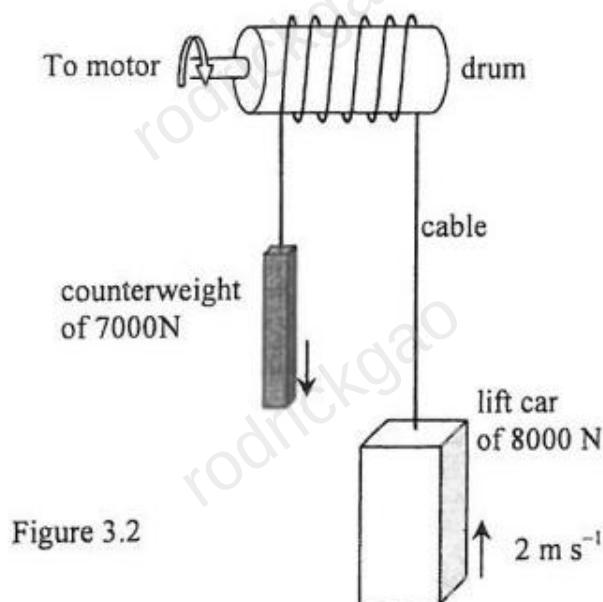


Figure 3.2

- (b)(i) Find the required total mechanical motor output, assuming the same frictional power loss as in (a). (2 marks)
- (b)(ii) State the advantage of the counterweight. (1 mark)

- (b)(iii) Comment on the claim that a frictionless drum surface would further reduce the motor power required. (2 marks)

Official marking scheme

3.	(a)	(i)	$P = F v$ $= 8000 \text{ N} \times 2 \text{ m s}^{-1}$ $= 16 \text{ kW}$	1M	
				1A	<u>2</u>
		(ii)	$P_{\text{loss}} = 20 \text{ kW} - 16 \text{ kW} = 4 \text{ kW}$	1A	<u>1</u>
	(b)	(i)	$P = 4 \text{ kW} + (8000 - 7000 \text{ N}) \times 2 \text{ m s}^{-1}$ $= 6000 \text{ W} = 6 \text{ kW}$	1M	
				1A	<u>2</u>
		(ii)	Output power required from motor is smaller. Or Force exerted by motor is smaller.	1A	<u>1</u>
		(iii)	No, the lift system could not work / the lift will fall as slipping would occur / the cable cannot be fixed on the drum / the drum cannot exert a force on the cable.	1A 1A	<u>2</u>

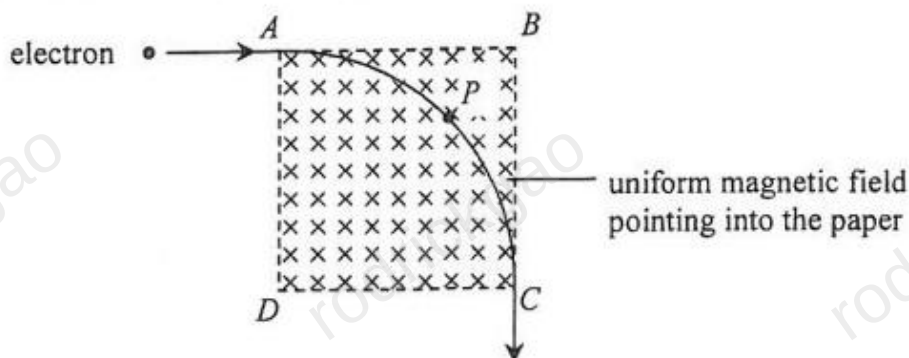
(a) Constant speed means the upward force on the car equals its weight. Useful mechanical power is $Fv = 16 \text{ kW}$; the difference between 20 kW and 16 kW is the 4 kW frictional loss.

(b)(i) The descending counterweight releases gravitational potential energy at $7000 \times 2 = 14 \text{ kW}$. The car gains 16 kW , so the motor need supply a net 2 kW plus the unchanged 4 kW loss: 6 kW .

(b)(ii–iii) A counterweight reduces the load on the motor. However, contact friction between cable and drum transmits the driving force; eliminating it makes the cable slip. Distinguish necessary static contact friction from dissipative losses in moving parts.

Question 4

* An electron of speed $1.2 \times 10^7 \text{ m s}^{-1}$ enters square ABCD, where a uniform 0.01 T magnetic field points into the paper. It follows a quarter circle from A to C and emerges with the same speed. Neglect gravity.



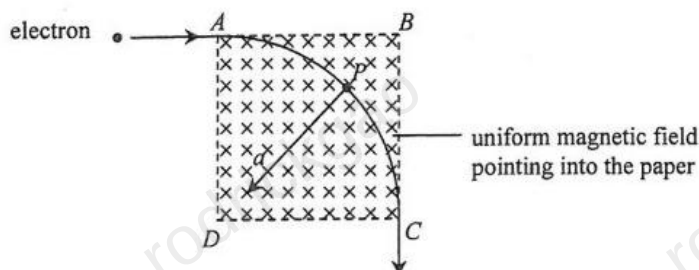
- (a)(i) Find the magnetic-force magnitude at P. (2 marks)
- (a)(ii) Indicate the acceleration direction at P on the diagram. (1 mark)
- (b) Although the electron accelerates, explain why its speed is unchanged on emerging. (2 marks)
- (c) Deduce the entry speed that would produce a semicircle from A to D instead. (2 marks)

Official marking scheme

4. (a) (i) $F = qvB = (1.60 \times 10^{-19} \text{ C})(1.2 \times 10^7 \text{ m s}^{-1})(0.01 \text{ T})$
 $= 1.92 \times 10^{-14} \text{ N}$

1M
1A 2

(ii)



1A 1

4. (b) F is perpendicular to the electron's displacement / velocity,
 Or electron only changes direction while speed remains unchanged,
 no work is done, therefore k.e. remains unchanged.

1A
1A 2

(c)
$$\left. \begin{aligned} F &= qvB = \frac{mv^2}{r} \\ \frac{v}{r} &= \frac{qB}{m} = \text{constant} \end{aligned} \right\}$$

1M
1A 2

v should be halved, i.e. $0.6 \times 10^7 \text{ m s}^{-1}$

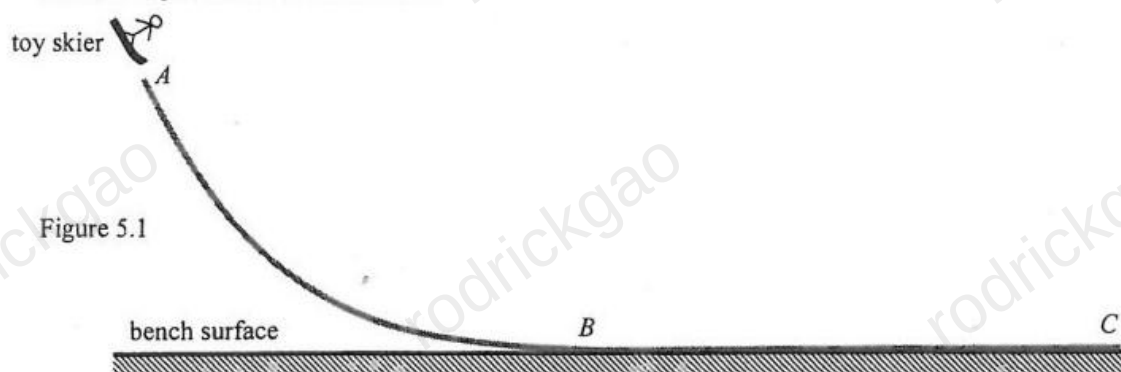
(a) Velocity is perpendicular to B, so use $|q|vB$ with the electron charge magnitude. At P the magnetic force and acceleration point radially inward towards the original circle centre D, not along the tangent.

(b) Acceleration means changing velocity, including direction. Here $F \cdot v = 0$, so magnetic force supplies no power and does not change kinetic energy or speed.

(c) The original quarter-circle radius is AD. A semicircle from A to D has diameter AD, hence radius AD/2. Since $r = mv/(|q|B)$, fixed B requires v to halve, giving $6.0 \times 10^6 \text{ m s}^{-1}$.

Question 5

A smooth sloping track ABC is fixed in a vertical plane, with horizontal part BC resting on a bench. You have a toy skier, a metre rule and a long rough paper strip with adhesive tape underneath.



Describe an experiment using this apparatus to investigate how stopping distance depends on release height. Include the quantities measured and the expected result. (5 marks)

Official marking scheme

5. Stick the paper strip onto the horizontal part *BC* of the track. 1A
 Release the toy from a certain height *h* from the bench surface and measure the corresponding stopping distance *d*. 1A
 Stopping distance should be measured from the beginning of the horizontal part *BC* or on the paper strip. 1A
 Release the toy from different heights and measure the corresponding stopping distances. 1A
 Plot a graph of *d* against *h*, 1A
 a straight line passing through the origin should be obtained. 1A
 or
 since $mgh = Fd$
 so $\frac{d}{h} = \text{constant} / d \propto h$ 5

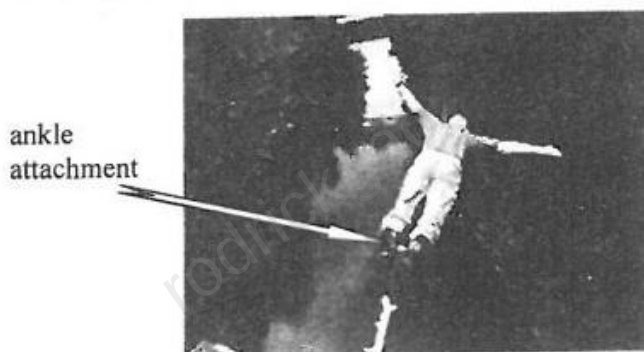
Attach the rough strip only to BC so the descent remains smooth. Measure vertical release height *h*, not distance along the slope. Release without pushing and measure *d* from where the toy first contacts the rough horizontal strip to where it stops.

Use several heights with the same toy and paper surface; repeat trials and average *d* to reduce scatter. Plot *d* vertically against *h* horizontally.

On the smooth slope, lost potential energy becomes kinetic energy. Approximately constant friction *F* on BC then does work $Fd = mgh$, so $d = (mg/F)h$: a straight line through the origin. The slope is not the acceleration.

Question 6

In bungee jumping a person jumps from a tall structure attached by a thick elastic cord. After some free fall the cord stretches; the jumper is momentarily at rest at the lowest point, then rebounds and oscillates before stopping.



A simple ankle attachment can secure the jumper, but some have detached, so many operators use a full body harness.



Neglect air resistance.

- (a)(i) Describe the acceleration during the first descent to the lowest point. (3 marks)
- (a)(ii) State the energy changes from the start until that lowest point. (2 marks)
- (b) In terms of net force, explain why the cord must be elastic. (2 marks)
- (c) In terms of contact area, explain why a full body harness is less likely to injure or detach than an ankle attachment. (2 marks)

Official marking scheme

- | | | | | | |
|----|-----|------|--|----------------|----------|
| 6. | (a) | (i) | Accelerates at g before the elastic cord stretches / at the beginning.
Acceleration decreases as the cord stretches.
Decelerates until momentarily at rest
(after the tension in the cord is greater than mg). | 1A
1A
1A | <u>3</u> |
| | | (ii) | Gravitational potential energy changed to kinetic energy and
(then) elastic potential energy in elastic cord. | 1A
1A | <u>2</u> |
| | (b) | | Elastic cord lengthens the stopping time,
hence reduces the (net) force acting on the player. | 1A
1A | <u>2</u> |
| | (c) | | Contact area is larger,
hence pressure is smaller during the fall and the structure is less likely to break / detach. | 1A
1A | <u>2</u> |

(a)(i) Take downward as positive: $a = g - T/m$. Before stretching $T = 0$; as T rises, downward acceleration falls to zero at $T = mg$, where downward speed is greatest. Afterwards $T > mg$ gives upward acceleration while the jumper still moves down and slows. At the lowest point $v = 0$ but acceleration is not zero.

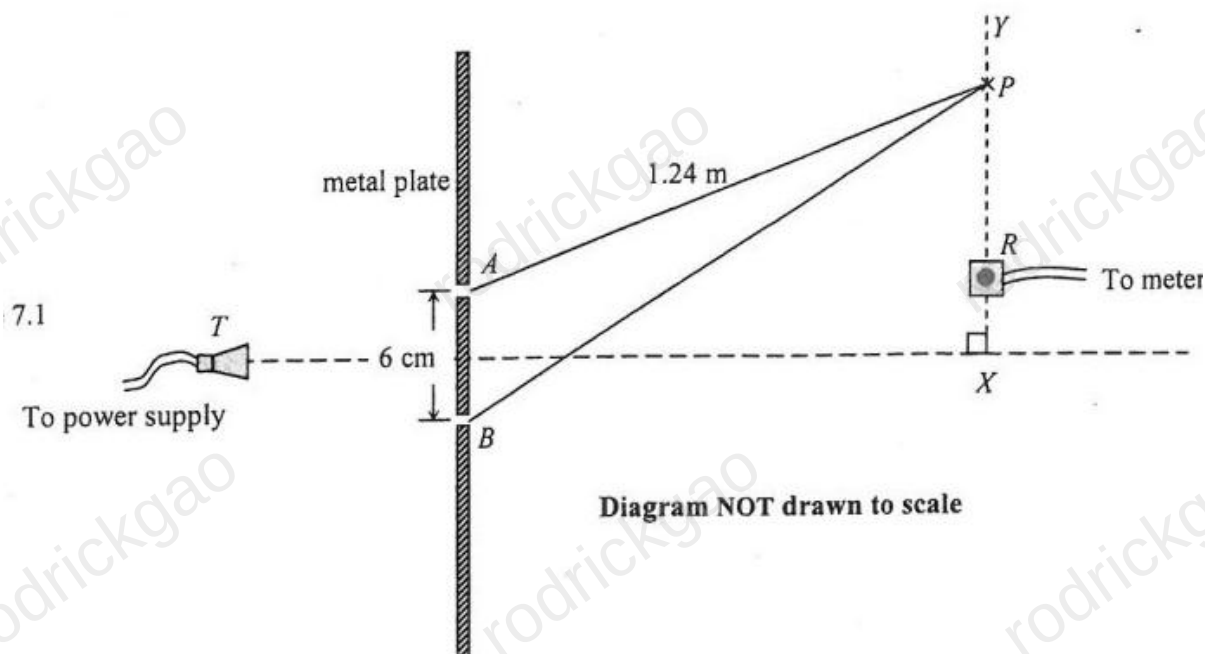
(a)(ii) Initially gravitational potential energy becomes kinetic energy. As the cord stretches, elastic potential energy builds while motion eventually slows; at the lowest point the kinetic energy is zero.

(b) For the same momentum change, increasing stopping time reduces average net force, $|F_{\text{net,avg}}| = |\Delta p|/\Delta t$. Elastic extension avoids an abrupt stop.

(c) For a comparable supporting force, pressure is F/A . Spreading it over the torso increases contact area and reduces concentrated pressure compared with a narrow ankle attachment.

Question 7

A microwave transmitter T emits waves of wavelength 2 cm towards two slits A and B , 6 cm apart in a metal plate. T and X are each equidistant from A and B . A receiver R moves along XY . At P , the second minimum is detected and $AP = 1.24$ m. The diagram is not to scale.



- (a) Find the microwave frequency. (2 marks)
- (b)(i) Explain why alternating maxima and minima are detected along XY . (2 marks)
- (b)(ii) Calculate BP . (2 marks)
- (b)(iii) As R moves from X towards Y and beyond, can more than three maxima be detected? Explain. (2 marks)
- (c) Microwaves are used in radar. Explain why radio waves of lower frequency are not suitable for this purpose. (2 marks)

Official marking scheme

- | | | | |
|---------|---|----------|----------|
| 7. (a) | $c = f\lambda \Rightarrow 3 \times 10^8 \text{ m s}^{-1} = f(0.02 \text{ m})$ | 1M | |
| | $\therefore f = 1.5 \times 10^{10} \text{ Hz or } 15000 \text{ MHz}$ | 1A | <u>2</u> |
| (b) (i) | Path difference of the diffracted waves from slits A and B to probe varies along XY .
Constructive and destructive interference occur alternately to give maxima and minima. | 1A
1A | <u>2</u> |
| (ii) | $BP - AP = 1\frac{1}{2}\lambda$
$BP - AP = 3 \text{ cm} = 0.03 \text{ m}$
$\therefore BP = 1.24 + 0.03 = 1.27 \text{ m}$ | 1M
1A | <u>2</u> |
| (iii) | Path difference along $XY < AB$
$AB = 3 \times 2 \text{ cm} = 3\lambda$
$\therefore$ path difference allowed $= 0\lambda, 1\lambda, 2\lambda$.
Maximum number of maxima $= 3$ | 1M
1A | <u>2</u> |
| (c) | Radio waves with lower frequencies (will have longer wavelengths and hence) have greater diffraction effect.
Radio waves by-pass small obstacles / not to be reflected from small obstacles. | 1A
1A | <u>2</u> |

(a) Convert 2 cm to 0.02 m before using $f = c/\lambda$.

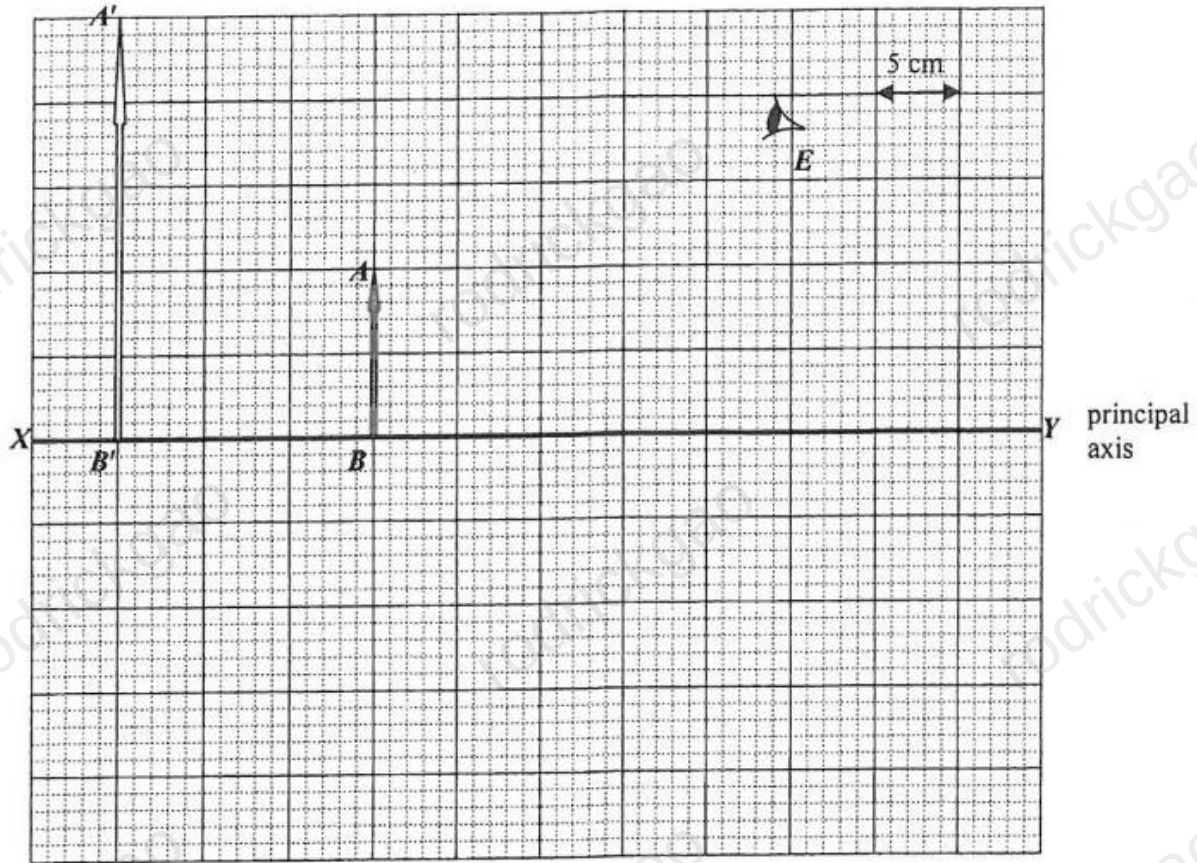
(b) T supplies the two slits in phase. Maxima occur at $\Delta = n\lambda$; minima occur at $\Delta = (n+\frac{1}{2})\lambda$. The second minimum has $\Delta = 3\lambda/2$, not 2λ . P is nearer A, hence $BP = AP + \Delta$.

(b)(iii) Include the central maximum at X. At every finite receiver position on this line, $|BP - AP| < AB = 6$ cm. A path difference of 3λ is only approached in the distant limiting direction, so it does not give a fourth finite-position maximum.

(c) In the small-target context of this question, a longer wavelength diffracts more around the target, weakening a useful reflected signal. This does not mean that every lower-frequency radar application is impossible.

Question 8

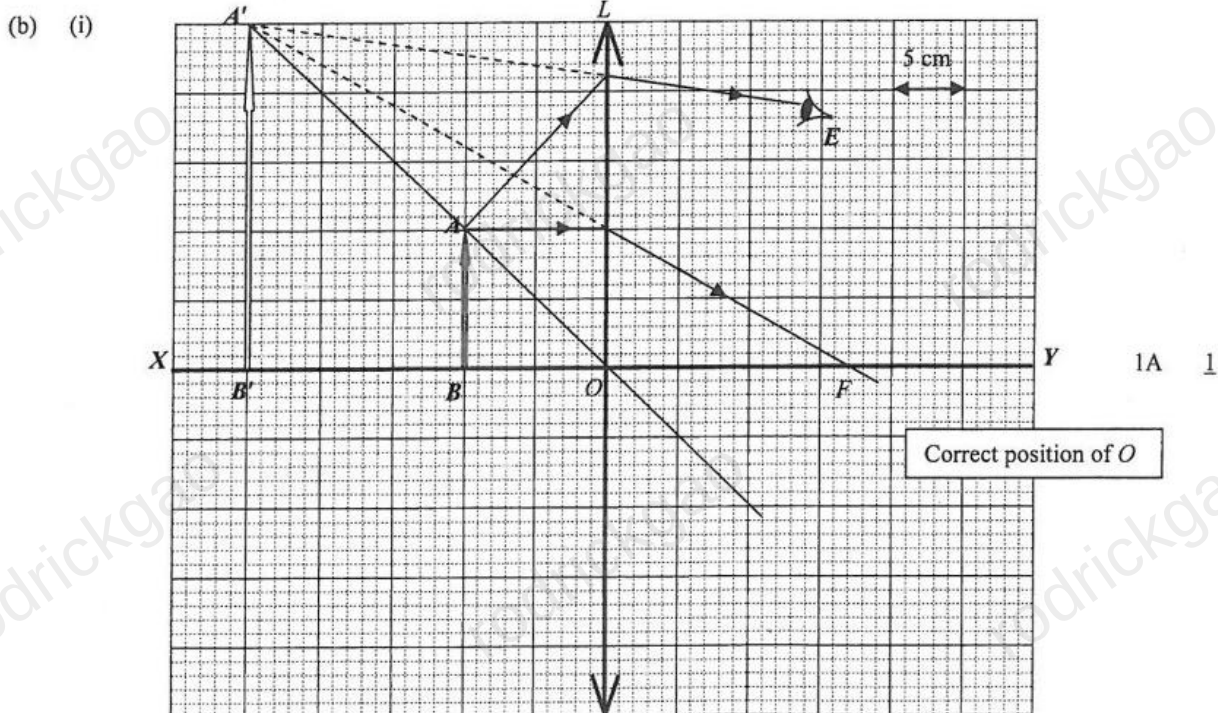
The grid shows an object AB, its image A'B' formed by a lens L (not shown), the principal axis XY and an observer's eye E. The horizontal scale is 1 cm on the original diagram representing 5 cm.



- (a)(i) Is A'B' real or virtual? (1 mark)
- (a)(ii) Identify the lens type and explain. (2 marks)
- (b)(i) Locate the optical centre O and draw lens L. (1 mark)
- (b)(ii) Draw an additional ray to locate the principal focus F and find the focal length. (2 marks)
- (c) Draw a light ray to show how E sees the image of the object's head A. (2 marks)
- (d) Give one application of this lens arrangement. (1 mark)

Official marking scheme

8. (a) (i) Virtual 1A 1
 (ii) Convex. 1A
 Only convex lens can form magnified (virtual, erect) images. 1A 2



- (ii) Correct light ray to locate F . 1M
 Focal length $f = 17$ cm (16.0 to 17.5 cm) 1A 2
 (c) Correct ray from A' or lens to E . 1A
 All correct. 1A 2
 (d) Magnifying glass / glasses for long-sighted eyes / simple microscope 1A 1

(a) The image is upright and enlarged. A diverging lens gives a diminished upright image, so this is a converging lens with the object inside its focal length.

(b) Join A' to A and extend to XY : a ray through the optical centre is undeviated, so that intersection is O . Draw the lens perpendicular to XY through O . A ray from A parallel to XY refracts through the far focus; its backward extension passes through A' .

Use the printed grid and stated scale, not a ruler on a resized screen. The object and virtual-image distances are approximately 10 cm and 25 cm: $1/f = 1/10 - 1/25$, giving $f \approx 16.7$ cm, consistent with the accepted graphical range.

(c) Join A' to E to locate its intersection with the lens. The actual ray travels $A \rightarrow \text{lens} \rightarrow E$. Only the backward extension runs to A' ; the virtual image does not emit light. The original marking diagram above shows the completed construction.

(d) A magnifying glass is one application: a convex lens with the object inside its focal length produces an enlarged upright virtual image.

Question 9

Carbon-14 decays by beta emission. An archaeological wood sample has activity 0.2 Bq. The half-life of carbon-14 is 5730 years. Take $1 \text{ year} = 3.16 \times 10^7 \text{ s}$.

*(a) Calculate the decay constant k in s^{-1} , then the number of carbon-14 nuclei remaining. (3 marks)

Living plants maintain a carbon-14 to carbon-12 ratio of 1.3×10^{-12} through carbon dioxide exchange.

(b) The sample contains 1×10^{23} carbon nuclei in total. Estimate the number of carbon-14 nuclei when the plant died. (1 mark)

*(c) Estimate the age of the wood in years. (2 marks)

Official marking scheme

9. (a) $k = \frac{\ln 2}{5730 \times 3.16 \times 10^7} = 3.83 \times 10^{-12} (\text{s}^{-1})$ 1A
- Activity $A = kN$
- $N = \frac{A}{k} = \frac{0.2}{3.83 \times 10^{-12}} = 5.22 \times 10^{10}$ 1M
1A 3
- (b) No. of ^{14}C nuclei: $N_0 = 1 \times 10^{23} \times (1.3 \times 10^{-12}) = 1.3 \times 10^{11}$ 1A 1
- (c) $kt = \ln \frac{N_0}{N}$
- $(3.83 \times 10^{-12}) t = \ln \frac{1.3 \times 10^{11}}{5.2 \times 10^{10}}$ 1M
- $t = 2.4 \times 10^{11} \text{ s}$ or 7571 (years) (accept 7500 to 7600 (years)) 1A 2

(a) Activity counts decays per second, not remaining nuclei. Convert the half-life to seconds to obtain k , then divide the measured activity by k .

(b) Carbon-14 is a tiny fraction of all carbon, so the total carbon count is approximately the carbon-12 count. Multiplying by the living ratio estimates N_0 .

(c) After death, replenishment stops and $N = N_0 e^{-kt}$. Hence $t = \ln(N_0/N)/k$, followed by division by 3.16×10^7 to convert seconds to years. The result is about 7.5×10^3 years. Intermediate rounding explains the official 7571-year value; do not treat its last digits as measurement precision.

Question 10

(a) A 12 V battery of negligible internal resistance is connected to a thermistor R and a $120\ \Omega$ resistor. The graph shows the thermistor resistance against temperature.

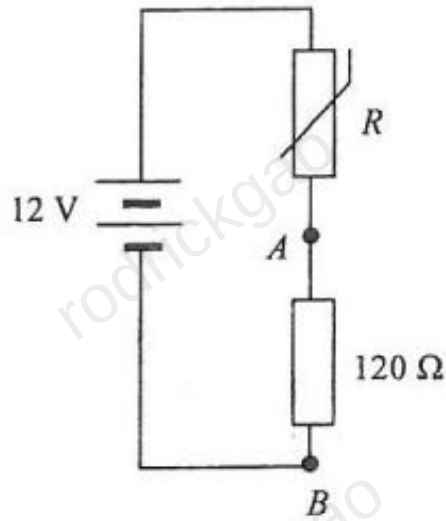
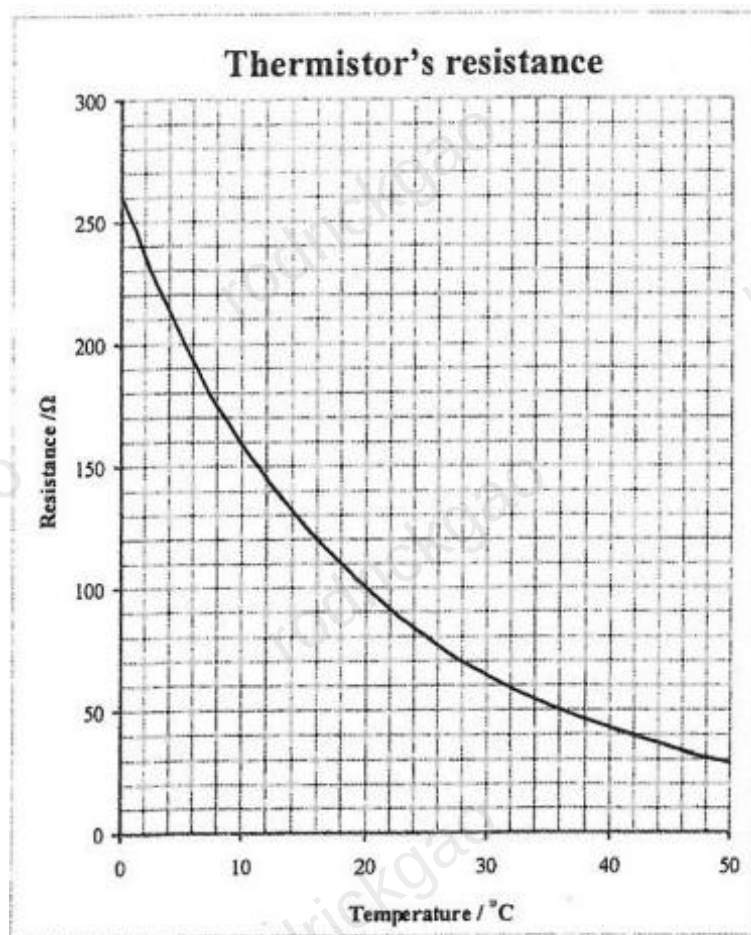


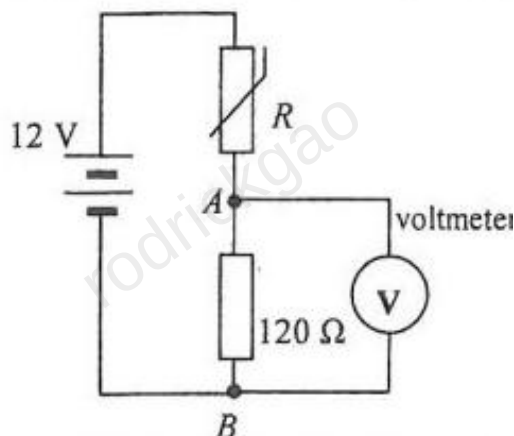
Figure 10.1



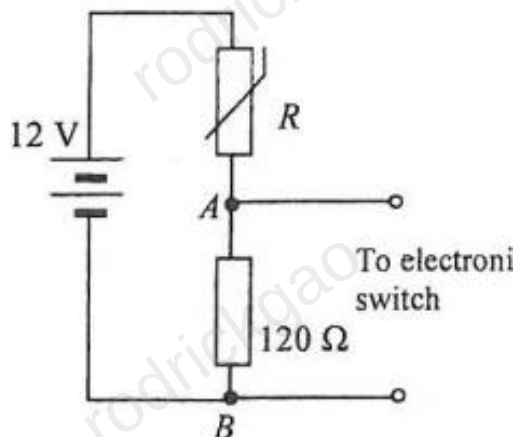
(a)(i) Find R at 25°C . (1 mark)

(a)(ii) Find the potential difference V_{AB} at 25°C . (2 marks)

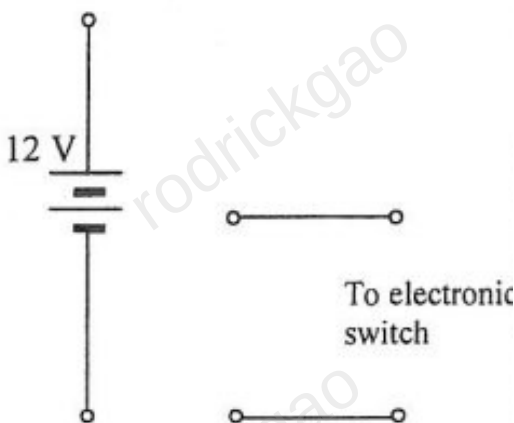
(b) Kelly measures V_{AB} with a voltmeter of about $1\text{ k}\Omega$ resistance. Despite careful measurement, its reading differs slightly from (a). Explain why and suggest how to improve accuracy. (3 marks)



(c)(i) V_{AB} drives an electronic switch connected across AB , turning a fan on when V_{AB} is 6.0 V or above. Use the graph to find the minimum temperature needed to keep the fan on. Show your working. (2 marks)



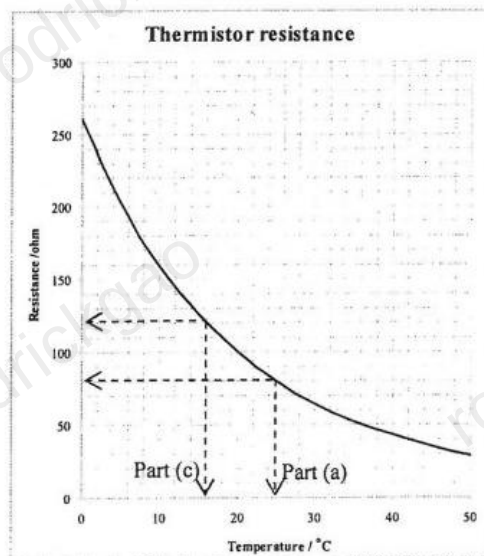
(c)(ii) Without additional components, complete the new circuit to turn on a heating device when temperature falls below a certain value. Explain the action; no calculation is required. (3 marks)



Official marking scheme

10. (a) (i) $80\ \Omega$

1A 1



$$(ii) \quad V_{AB} = \frac{120}{(80 + 120)} \times 12 = 7.2\text{ V}$$

1M

1A 2

- (b) As R_v and $120\ \Omega$ resistor are in parallel, R_{eq} across AB is smaller than $120\ \Omega$, therefore voltage shared across AB is reduced / smaller than expected.

1A

1A

Use a voltmeter with resistance much larger than the resistance in that part of the circuit. (e.g. $10\text{ M}\Omega$ in some digital voltmeter)

1A

3

$$(c) (i) \quad V_{AB} = \frac{120}{(R + 120)} \times 12 = 6.0\text{ V}$$

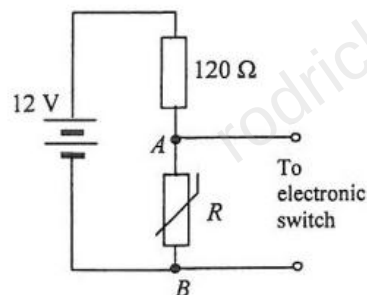
$$R = 120\ \Omega$$

1A

corresponds to temperature at 16°C .

1A 2

10. (c) (ii) Correct circuit (i.e. interchange thermistor R and $120\ \Omega$ resistor).
As the temperature drops, the thermistor resistance increases.
When the resistance increases to a value such that $V_{AB} = 6.0\text{ V}$ or above, the electronic switch is on and it turns on the heating device.



1A

1A

1A 3

(a) Read $80\ \Omega$ at 25°C . The series current is $12/(80+120) = 0.060\text{ A}$; across the fixed resistor, $V_{AB} = 0.060 \times 120 = 7.2\text{ V}$.

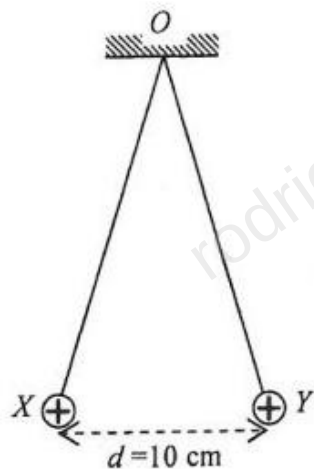
(b) A real voltmeter draws current. For $1\text{ k}\Omega$, $R_{\text{parallel}} = (120 \times 1000)/(120 + 1000) \approx 107\ \Omega$, giving a loaded reading about 6.87 V . A high-resistance meter makes this loading effect much smaller; repeated careful reading alone cannot remove it.

(c)(i) At the switch threshold the fixed resistor receives half the supply, so both series resistances are $120\ \Omega$. The graph gives 16°C . Warmer temperatures reduce R and increase V_{AB} , keeping the fan on.

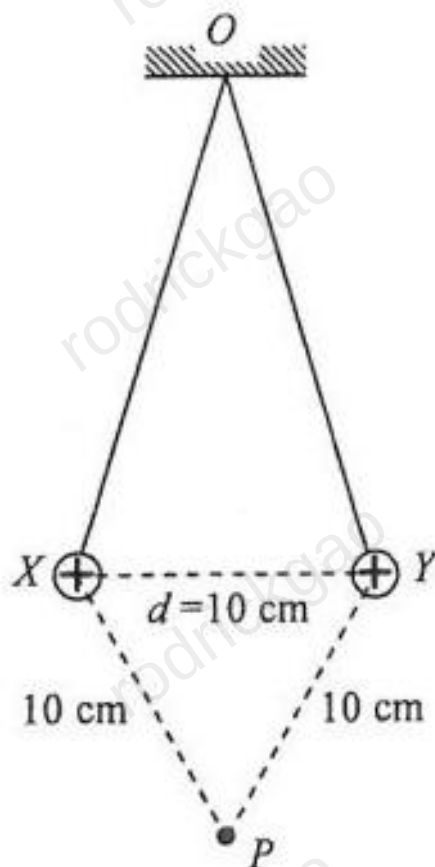
(c)(ii) The original marking diagram above places the thermistor in the lower branch across AB . Now $V_{AB} = 12R/(120+R)$, which increases when cooling increases R . The same high-voltage trigger therefore operates a heater on cooling.

Question 11

Two identical small metal spheres X and Y hang from equal-length insulating threads. Each sphere has mass 1.0×10^{-5} kg and charge $+3.1$ nC ($1 \text{ nC} = 10^{-9} \text{ C}$). Their separation d is 10 cm. Their sizes are negligible compared with d , so treat them as point charges. Take $1/(4\pi\epsilon_0) = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$ and $g = 9.81 \text{ m s}^{-2}$. The diagram is not to scale.



- (a) Find the angle between the threads. (3 marks)
- (b) P is vertically below O and 10 cm from each sphere.



- (b)(i) Indicate the direction of the resultant electric field at P due to the two charged spheres. (1 mark)
- * (b)(ii) Calculate the electric potential at P, taking the potential at infinity as zero. (2 marks)

(b)(iii) A neutral metal sphere of finite size is placed at P. State whether d increases, decreases or stays unchanged. (1 mark)

Official marking scheme

11. (a)

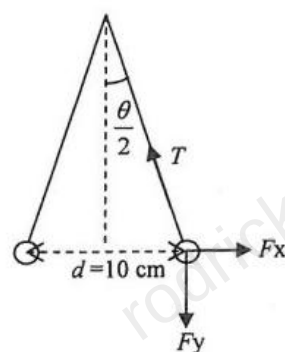
$$T \cos \frac{\theta}{2} = F_y = mg$$

$$T \sin \frac{\theta}{2} = F_x = \frac{Q^2}{4\pi \epsilon_0 d^2}$$

$$\tan \frac{\theta}{2} = \frac{Q^2}{4\pi \epsilon_0 d^2} \left(\frac{1}{mg} \right)$$

$$= 9 \times 10^9 \times \frac{(3.1 \times 10^{-9})^2}{0.1^2} \times \frac{1}{(1.0 \times 10^{-5})(9.81)}$$

$$\frac{\theta}{2} = 5.0^\circ \text{ i.e. } \theta = 10.1^\circ$$

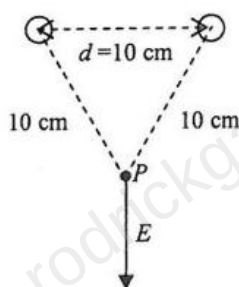


1M

1M

1A 3

(b) (i)



1A 1

(ii) Potential at P $= \frac{Q}{4\pi \epsilon_0 d} + \frac{Q}{4\pi \epsilon_0 d} = \frac{2Q}{4\pi \epsilon_0 d}$

1M

$$= (9 \times 10^9) \frac{2 \times 3.1 \times 10^{-9}}{0.1}$$

$$= 558 \text{ V}$$

1A 2

(iii) Separation d decreases.

1A 1

(a) Each sphere is in equilibrium under tension, weight and horizontal electric repulsion. Let θ be the full angle between the threads, so each thread is $\theta/2$ from the vertical, as in the original marking diagram. $F_e = kQ^2/d^2 = 8.649 \times 10^{-6} \text{ N}$ and $mg = 9.81 \times 10^{-5} \text{ N}$. Therefore $\theta = 2 \arctan(F_e/mg) \approx 10.1^\circ$, not 5.0° .

(b)(i) Electric fields point away from positive charges. At P their horizontal components cancel, while their downward components add.

(b)(ii) Electric potential is a scalar. Each charge contributes +279 V, so $V = 558 \text{ V}$; do not resolve potentials into components.

(b)(iii) The neutral conducting sphere polarizes. Induced charge makes it attract the positive spheres towards the central region; their equilibrium separation becomes smaller. Neutral does not mean unable to experience electrostatic attraction.